

Problem Set 2

Fiscal and Monetary Policy, Spring 2026

June 1, 2026

This problem set consists of two problems. Submit your solutions until **June 16th, 11:59 PM**. You can work in teams of up to two students.

Problem 1: Optimal policy

Part A: Lucas–Stokey with two government-spending states.

Time is discrete and infinite. At each date the exogenous state is $s_t \in \{L, H\}$, where government purchases are

$$g_L < g_H.$$

The state follows a Markov chain with transition matrix

$$\Pi = \begin{bmatrix} \pi_{LL} & \pi_{LH} \\ \pi_{HL} & \pi_{HH} \end{bmatrix}.$$

The representative household has preferences

Rows index today's state and columns index tomorrow's state.

$$E_0 \sum_{t=0}^{\infty} \beta^t (\log c(s^t) - \psi \ell(s^t)), \quad 0 < \beta < 1, \quad \psi > 0,$$

where $\ell(s^t)$ is labor. Output is produced one-for-one from labor:

$$y(s^t) = \ell(s^t).$$

The government levies a proportional labor income tax τ_t and can trade a full set of one-period Arrow securities. Let $a(s^t)$ denote the household's beginning-of-period asset position at history s^t , and let $q(s^{t+1} | s^t)$ denote the price at history s^t of a claim paying one unit of goods next period if successor history s^{t+1} occurs.

The household's sequential budget constraint is

$$c(s^t) + \sum_{s^{t+1} \geq s^t} q(s^{t+1} | s^t) a(s^{t+1}) = (1 - \tau(s^t)) \ell(s^t) + a(s^t),$$

with initial asset position $a(s^0) = a_0 = 0$. The resource constraint is

$$c(s^t) + g(s_t) = \ell(s^t).$$

1. Derive the household optimality conditions, Arrow-security prices, and the labor tax wedge implied by the intratemporal condition. Explain how these conditions are used in the primal approach to eliminate prices and taxes.

2. Define the pseudo-utility function

$$W(c, \ell; \Phi) \equiv \log c - \psi \ell + \Phi (u_c(c, \ell)c + u_\ell(c, \ell)\ell),$$

where Φ is the implementability constraint multiplier. Derive consumption, labor, and the tax rate under optimal policy in each state. Explain why the tax rate is constant even though government purchases are stochastic.

3. Let primary surpluses be $z(s) \equiv \tau(s)\ell(s) - g(s)$. Let $b(s)$ denote the inherited Arrow obligation due in state s , and define $x(s) \equiv u_c(s)b(s)$. Derive the Markov system that determines $\vec{x}$ and recover $b(s)$ in each state. Interpret the cross-state pattern of inherited debt.
4. For the numerical part, use the following primitive calibration:

$$\beta = 0.95, \quad \psi = 1, \quad g_L = 0.10, \quad g_H = 0.30,$$

$$\Pi = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}.$$

Assume the initial state is $s_0 = L$. Compute the Ramsey allocation, tax rate, primary surpluses, Arrow price matrix, and inherited Arrow payoffs.

Part B: Implementing the state-contingent debt burden with a bill and a consol.

We will now try to implement the optimal allocation using a more limited set of securities.

Suppose that at date $t - 1$ the government can choose two noncontingent real positions:

$$B_{t-1}^1 \text{ one-period bills, } B_{t-1}^2 \text{ consols.}$$

Positive B means the government has issued debt; negative B means the government holds the security as an asset. A one-period bill pays one unit of goods at date t in every state. A consol pays one unit of goods every period forever. Let $p_t^C(s_t)$ denote the date- t market value of one inherited consol after state $s_t \in \{L, H\}$ is realized, including the coupon due at date t .

Use the same transition matrix as in Part A. Since the Part A allocation has constant consumption, the one-period real discount factor is β in both states. Since $p_t^C(s_t)$ includes the current coupon, it satisfies the recursive pricing equation

$$p_t^C(s) = 1 + \beta \sum_{s' \in \{L, H\}} \pi(s' | s) p_{t+1}^C(s').$$

Keep the sign convention fixed: positive B is a government liability. The price p_t^C includes the coupon due at date t .

The date- t market value of the inherited maturity portfolio is therefore

$$V_t^M(s_t) = B_{t-1}^1 + p_t^C(s_t)B_{t-1}^2.$$

5. Derive the two consol values $p_t^C(L)$ and $p_t^C(H)$ implied by the Part A allocation and the calibration above. Explain why this simple Lucas–Stokey example does or does not generate state-dependent long-bond values.
6. Use the inherited Arrow payoffs from Part A as the complete-markets target $b_t^A(s_t)$. Set up the system $V_t^M(s_t) = b_t^A(s_t)$ and determine whether it has a solution. Interpret the result. Link it to the following: derive the spanning condition for implementing arbitrary two-state debt burdens with one bill and one consol. Apply it to the matrix

$$A_t = \begin{bmatrix} 1 & p_t^C(H) \\ 1 & p_t^C(L) \end{bmatrix}$$

and explain what the condition requires economically.

7. Let's move away from the particular setting analyzed above. Suppose the richer model instead implies consol values:

$$p_t^C(H) = 16, \quad p_t^C(L) = 20,$$

and suppose the target state-contingent debt burdens are

$$b_t^A(H) = 0.30, \quad b_t^A(L) = 0.50.$$

Derive the bill-consol portfolio that implements these debt burdens and interpret the signs of the two positions.

8. Suppose the richer model instead implies consol values:

$$p_t^C(H) = 19, \quad p_t^C(L) = 20,$$

while the target remains $b_t^A(H) = 0.30$, $b_t^A(L) = 0.50$. Derive the new bill-consol portfolio and relate the result to Buera–Nicolini's point about nearly singular return matrices.

Problem 2: FTPL with long-term nominal debt and surplus shocks

This problem extends the fiscal theory of the price level to an economy with long-term nominal debt.

Consider a cashless endowment economy with constant real endowment. The representative household has stochastic discount factor β , where $0 < \beta < 1$. Fiscal policy is active: the real primary-surplus process is fixed independently of the price level.

At the beginning of date t , the government has inherited two nominal liabilities:

B_{t-1}^1 one-period nominal debt due at date t ,

and

B_{t-1}^C nominal consols.

A nominal consol pays one unit of currency every period forever. Let Q_t^{ex} be the ex-coupon nominal price of the consol at date t . Thus the coupon-inclusive market value at date t of one inherited consol is

$$M_t \equiv 1 + Q_t^{ex}.$$

Let S_t denote the real primary surplus. Define fiscal backing by

$$\mathcal{F}_t \equiv \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j S_{t+j}.$$

In this problem we will study what happens in the aftermath of a fiscal shocks. Treat the post-shock surplus and price-level paths as deterministic conditional on date-0 information.

1. General valuation formula.

At date t , derive the FTPL valuation equation linking inherited nominal liabilities, the price level, and fiscal backing:

$$\frac{B_{t-1}^1 + M_t B_{t-1}^C}{P_t} = \mathcal{F}_t.$$

Next, derive the coupon-inclusive nominal value of one inherited consol:

$$M_t = \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \frac{P_t}{P_{t+j}}.$$

x For the deterministic paths considered below, this becomes

$$M_t = \sum_{j=0}^{\infty} \beta^j \frac{P_t}{P_{t+j}}.$$

Keep three objects separate: the current price level P_t , the expected future price-level path, and the market value of inherited long bonds.

Fiscal backing is measured in goods. The left-hand side of the FTPL equation converts nominal liabilities into goods by dividing by P_t .

When paths are stochastic, put the relevant conditional expectation in front of future-price sums.

Timing: B_{t-1}^1 and B_{t-1}^C are inherited; P_t and M_t are date- t values.

Specialize the formula to date 0 and derive

$$\frac{B_{-1}^1}{P_0} + B_{-1}^C \sum_{j=0}^{\infty} \beta^j \frac{1}{P_j} = \mathcal{F}_0.$$

I want to see your derivations. Compare this formula with the one-period-debt case. Explain why one-period nominal debt pins down P_0 directly, while long-term nominal debt pins down a restriction on the whole expected path of price levels.

2. **A surplus shock.**

Suppose that after a date-0 fiscal shock, the primary-surplus sequence is deterministic and given by

$$S_j = \bar{S} - \Delta \rho^j, \quad 0 < \rho < 1, \quad \Delta > 0.$$

Assume

$$\frac{\bar{S}}{1 - \beta} - \frac{\Delta}{1 - \beta \rho} > 0.$$

Derive $\mathcal{F}_0$. Then specialize to

$$\beta = 0.95, \quad \bar{S} = 5, \quad \rho = 0.8, \quad \Delta = 4.8.$$

Compute $\mathcal{F}_0$. Also compute $\mathcal{F}_t$ for general t and report $\mathcal{F}_1$ and $\mathcal{F}_2$.

3. **One-period nominal debt: impact revaluation and post-shock paths.**

Suppose first that the government enters date 0 with only one-period nominal debt:

$$B_{-1}^1 = 100, \quad B_{-1}^C = 0.$$

The surplus shock from the previous part lowers fiscal backing to

$$\mathcal{F}_0 = 80.$$

The date-0 FTPL valuation equation is therefore

$$\frac{B_{-1}^1}{P_0} = \mathcal{F}_0.$$

- (a) Compute the date-0 price level P_0 , assuming the pre-shock price level was $P_{-1} = 1$. Interpret this as expected or unexpected inflation.

Timing hint: P_{-1} is pre-shock, while P_0 is chosen after the fiscal shock is known.

- (b) Let q_t^n denote the date- t nominal price of a one-period nominal bond paying one unit of currency at date $t + 1$. Under the deterministic paths considered here, the nominal bond-pricing equation is

$$q_t^n = \beta \frac{P_t}{P_{t+1}}.$$

Explain why this equation determines expected inflation from date t to date $t + 1$, but does not prevent the date-0 revaluation of inherited nominal debt.

- (c) Now consider alternative arrangements from date 0 onward.

Arrangement A: fixed future one-period debt. Suppose the government rolls over one-period nominal debt so that the face value due at every future date is fixed:

$$B_{t-1}^1 = D = 100, \quad t \geq 0.$$

Use

$$\frac{D}{P_t} = \mathcal{F}_t$$

to derive P_t , $\Gamma_{t+1} \equiv P_{t+1}/P_t$, and q_t^n . Compute P_1 , P_2 , Γ_1 , Γ_2 , q_0^n , and q_1^n .

Arrangement B: zero expected inflation after date 0. Suppose monetary policy sets

$$q_t^n = \beta, \quad t \geq 0,$$

so that

$$P_{t+1} = P_t.$$

Starting from the date-0 price level P_0 , derive the future nominal debt face values B_t^1 required to satisfy the FTPL valuation equation at dates $t + 1$. Compute B_0^1 and B_1^1 .

Arrangement C: constant expected gross inflation after date 0. More generally, suppose monetary policy sets a constant expected gross inflation rate

$$\Gamma \equiv \frac{P_{t+1}}{P_t}, \quad t \geq 0.$$

Equivalently,

$$q_t^n = \frac{\beta}{\Gamma}.$$

Starting from P_0 , derive the implied price-level path P_t and the future nominal debt face values B_{t-1}^1 required to satisfy the FTPL valuation equation.

Separate the impact jump from -1 to 0 from expected inflation between 0 and 1 .

4. **Long-term nominal debt and the role of expected inflation.**

Now suppose the government has only nominal consols:

$$B_{-1}^1 = 0, \quad B_{-1}^C = 5.$$

Using the bad-surplus-shock value of $\mathcal{F}_0$ from part 2, derive the date-0 valuation restriction linking M_0 and P_0 .

Do not choose P_0 and M_0 independently; the valuation equation links them.

5. **Constant expected inflation.**

Suppose that from date 0 onward, expected gross inflation is constant:

$$P_j = P_0 \Gamma^j, \quad j = 0, 1, 2, \dots,$$

with $\Gamma > \beta$.

First derive the coupon-inclusive consol value

$$M_0(\Gamma) = \frac{\Gamma}{\Gamma - \beta}.$$

Then use the valuation equation to solve for

$$P_0(\Gamma) = \frac{B_{-1}^C}{\mathcal{F}_0} \frac{\Gamma}{\Gamma - \beta}.$$

Use

$$\beta = 0.95, \quad B_{-1}^C = 5, \quad \mathcal{F}_0 = 80$$

to compute P_0 under zero expected inflation, $\Gamma = 1$.

Next, solve for the permanent expected gross inflation rate Γ that keeps the current price level fixed at

$$P_0 = 1.$$

Interpret why a small permanent increase in expected inflation has a large effect on the value of long-term nominal debt.

6. **Active versus passive fiscal policy.**

In the calculations above, fiscal policy is active: the surplus path $\{S_t\}$, and therefore $\mathcal{F}_t$, is fixed independently of the price level. Suppose instead that fiscal policy is passive and future surpluses adjust so that

$$\mathcal{F}_t = \frac{B_{t-1}^1 + M_t B_{t-1}^C}{P_t}$$

for whatever price-level path is selected by monetary policy.

Does the FTPL valuation equation determine the price level in this case? Explain.

Which object is exogenous in this part, $\mathcal{F}_t$ or the price-level path?

7. Discussion.

Does long-term nominal debt make fiscal inflation disappear? Or does it change the timing of inflation and the asset prices through which the fiscal shock is absorbed? What is the role of monetary policy? Use your formulas and numerical answers above to support your argument.